

Chapter 6

Translation of Pólya and Schur's 1914 Paper

This chapter contains an English translation of Pólya and Schur's work titled *Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen*¹, which we will refer to as **PS**. The paper was published in 1914 in issue 144 of the *Journal für die reine und angewandte Mathematik* (commonly known as Crelle's journal, named after its founder). The same issue also includes Schur's solo paper, *Zwei Sätze über algebraische Gleichungen mit lauter reellen Wurzeln*², which appears immediately before **PS**. We will refer to Schur's solo paper as **S**.

The two papers, **S** and **PS**, can be viewed as a single work, as **PS** uses the first theorem of **S**. The fact that both papers were published in the same issue, with **S** preceding **PS**, supports this view.

However, due to time constraints, we were unable to translate **S**. Nonetheless, the first theorem of **S**, which is required by **PS**, has been fully proved in §2.4 of this thesis. In **PS**, this theorem is referred to as 'Schur's composition theorem' and **S** is simply referred to as 'A.' Since **PS** only depends on Schur's composition theorem, which we have already proved in this thesis, readers will be in a position to completely understand **PS** after reviewing the proof of Schur's composition theorem from Chapter 2.

The papers [2] and [1] cite results from **PS**, but we did not come across any existing English translations or comprehensive expositions of its content.

Our translation stays as close as possible to the original text, often rendering lines almost literally. We take full responsibility for any errors that may have arisen, particularly given our limited proficiency in German.

¹On two kinds of multiplier sequences in the theory of algebraic equations

²Two theorems about algebraic equations with real roots

On Two Kinds of Multiplier Sequences in the Theory of Algebraic Equations

G. Pólya and J. Schur

1914

Laguerre gave many examples of real sequences

$$\alpha_0, \alpha_1, \alpha_2, \dots \tag{A}$$

which have the following curious property: if

$$a_0 + a_1x + a_2x^2 + \dots + a_mx^m = 0$$

is a real polynomial with only real roots¹, then the roots of the polynomial

$$\alpha_0a_0 + \alpha_1a_1x + \alpha_2a_2x^2 + \dots + \alpha_ma_mx^m = 0$$

are all real. A sequence (A), with this property, is called a *multiplier sequence of the first kind*.

By a *multiplier sequence of the second kind*, we mean a sequence of real numbers

$$\beta_0, \beta_1, \beta_2, \dots \tag{B}$$

such that if

$$b_0 + b_1x + b_2x^2 + \dots + b_mx^m = 0$$

is a real polynomial with only reals *of the same sign*, then the polynomial

$$\beta_0b_0 + \beta_1b_1x + \beta_2b_2x^2 + \dots + \beta_mb_mx^m = 0$$

has only real roots. By a system of real numbers ‘of the same sign’, we mean that the non-zero members of the system are either all positive or all negative.

We mention a few of the examples given by Laguerre: the sequences

$$1, \frac{1}{\omega}, \frac{1}{\omega(\omega+1)}, \frac{1}{\omega(\omega+1)(\omega+2)}, \dots \quad (\omega > 0), \tag{1}$$

¹Here we consider the constant polynomials to have only real zeroes.

$$1, q, q^4, q^9, \dots \quad (|q| \leq 1) \quad (2)$$

are multiplier sequences of the first kind, furthermore, when λ and ϑ are real, the quantities

$$\cos(\lambda), \cos(\lambda + \vartheta), \cos(\lambda + 2\vartheta), \dots \quad (3)$$

form a multiplier sequence of the second kind.²

A multiplier sequence of the first kind is naturally also of the second kind. The converse is not always true. For example, the sequence (3), when ϑ is not an integer multiple of π , is not a multiplier sequence of the first kind.

In what follows, we will give necessary and sufficient criteria for each kind of multiplier sequences (§§ 3 and 4). In particular, we will present a correspondence between the multiplier sequences of both kinds and two classes of transcendental entire functions. This gives us a remarkable method of deriving old as well as new theorems about transcendental entire functions of genus 0 and 1 with all real zeroes (cf. § 6).

We expand upon the above work^a by Schur (which will be briefly cited as ‘A’ below), in which he proved his composition theorem.

1 Elementary properties of multiplier sequences

Theorem 1. *In a multiplier sequence of the first kind*

$$\alpha_0, \alpha_1, \alpha_2, \dots \quad (A)$$

the non-zero terms form a ‘continuous segment’ and either have the same sign or alternate.

Thus if α_λ and $\alpha_{\lambda+\mu}$ are non-zero, then none of the in-between terms vanish; additionally

$$\text{sign } \alpha_{\nu-1} = \text{sign } \alpha_{\nu+1}. \quad (\nu = \lambda + 1, \dots, \nu = \lambda + \mu + 1) \quad (4)$$

Let

$$a_\lambda x^\lambda + \dots + a_{\lambda+\mu} x^{\lambda+\mu} = 0$$

by any polynomial with only real roots, where a_λ and $a_{\lambda+\mu}$ are non-zero. By assumption, it follows that

$$\alpha_\lambda a_\lambda x^\lambda + \alpha_{\lambda+1} a_{\lambda+1} x^{\lambda+1} + \dots + \alpha_{\lambda+\mu} a_{\lambda+\mu} x^{\lambda+\mu} = 0$$

has only real roots. This implies that no two consecutive terms of the sequence

$$\alpha_\lambda, \alpha_{\lambda+1}, \dots, \alpha_{\lambda+\mu-1}, \alpha_{\lambda+\mu}$$

can vanish (cf. A, Proposition 1^b).

²*Oeuvres de Laguerre*, vol. I, pp. 31-35 and pp. 199-206.

But even the case that a term α_ν of this sequence is zero can be excluded. If $\alpha_\nu = 0$, then $\alpha_{\nu-1}$ and $\alpha_{\nu+1}$ have to be non-zero; furthermore, from the polynomials

$$x^{\nu-1} - x^{\nu+1} = 0 \text{ and } x^{\nu-1} + 2x^\nu + x^{\nu+1} = 0,$$

we obtain that each of the two polynomials

$$\alpha_{\nu-1} - \alpha_{\nu+1}x^2 = 0 \tag{5}$$

and

$$\alpha_{\nu-1} + \alpha_{\nu+1}x^2 = \alpha_{\nu-1} + 2\alpha_\nu x + \alpha_{\nu+1}x^2 = 0$$

have only real roots. But this is not possible.

Since (5) must have only real roots, regardless of the value of α_ν , (4) is true.

It is even easier to prove:

Theorem 2. *In a multiplier sequence of the second kind, if two adjacent terms are zero, then either all preceding or all following terms are zero. Further, if one term in the sequence is zero and the two adjacent terms are non-zero, then they must have opposite signs.*

Furthermore, we have:

Theorem 3. *In a multiplier sequence*

$$\gamma_0, \gamma_1, \gamma_2, \dots, \gamma_\nu, \dots$$

of either kind,

$$\gamma_\nu^2 - \gamma_{\nu-1}\gamma_{\nu+1} \geq 0^3.$$

We note that the equation

$$x^{\nu-1} + 2x^\nu + x^{\nu+1} = 0$$

has only real roots of the same sign, so it follows that the equation

$$\gamma_{\nu-1}x^{\nu-1} + 2\gamma_\nu x^\nu + \gamma_{\nu+1}x^{\nu+1} = 0$$

also has only real roots, and from this the proposition follows.

If

$$\alpha_0, \alpha_1, \alpha_2, \dots, \alpha_\nu, \dots \tag{A}$$

and

$$\beta_0, \beta_1, \beta_2, \dots, \beta_\nu, \dots, \tag{B}$$

are two sequences, then

$$\alpha_0\beta_0, \alpha_1\beta_1, \alpha_2\beta_2, \dots, \alpha_\nu\beta_\nu, \dots$$

is the sequence *composed* of (A) and (B). Then we have:

³Therefore, the sequence (2) from the introduction is not a multiplier sequence for $|q| > 1$.

Theorem 4. *The composition of two multiplier sequences of the first kind is a multiplier sequence of the first kind. A multiplier sequence of the first kind, composed with a multiplier sequence of the second kind, is a multiplier sequence of the second kind.*

The first part of this proposition is obvious. If (A) is a multiplier sequence of the first kind, (B) is a multiplier sequence of the second kind and

$$b_0 + b_1x + b_2x^2 + \cdots + b_mx^m = 0$$

is an equation with only real roots, then the equation

$$\alpha_0b_0 + \alpha_1b_1x + \alpha_2b_2x^2 + \cdots + \alpha_mb_mx^m = 0$$

has only real roots, which by Proposition 1 are of the same sign. Therefore, the equation

$$\beta_0\alpha_0b_0 + \beta_1\alpha_1b_1x + \beta_2\alpha_2b_2x^2 + \cdots + \beta_m\alpha_mb_mx^m = 0$$

has only real roots.

Theorem 5. *If*

$$\gamma_0, \gamma_1, \gamma_2, \dots, \gamma_\nu, \dots$$

is a multiplier sequence of either kind, then for any ν

$$\gamma_\nu, \gamma_{\nu+1}, \gamma_{\nu+2}, \dots$$

is a multiplier sequence of the same kind.

If the equation

$$c_0 + c_1x + \cdots + c_mx^m = 0$$

has only real roots, or has real roots of the same sign, then

$$c_0x^\nu + c_1x^{\nu+1} + \cdots + c_mx^{\nu+m} = 0$$

has the same property. Therefore

$$\gamma_\nu c_0x^\nu + \gamma_{\nu+1}c_1x^{\nu+1} + \cdots + \gamma_{\nu+m}c_mx^{\nu+m} = 0$$

or equivalently,

$$\gamma_\nu c_0 + \gamma_{\nu+1}c_1x + \cdots + \gamma_{\nu+m}c_mx^m = 0$$

has only real roots in both cases. Hence the proposition is proved.

Theorem 6. *Let, for each natural number n ,*

$$\gamma_{n0}, \gamma_{n1}, \gamma_{n2}, \dots$$

be a multiplier sequence of the first (or second) kind such that for each ν

$$\lim_{n \rightarrow \infty} \gamma_{n\nu} = \gamma_\nu.$$

Then

$$\gamma_0, \gamma_1, \gamma_2, \dots$$

is a multiplier sequence of the first (or second) kind.

This follows easily from Proposition 3^c of A.

2 On two classes of entire functions with real coefficients

An *entire function of type I* is an entire function

$$\Phi(x) = \alpha_0 + \frac{\alpha_1}{1!}x + \frac{\alpha_2}{2!}x^2 + \dots$$

with only real zeroes of the same sign, for which either $\Phi(x)$ or $\Phi(-x)$ can be written as a product of the form

$$\Phi(x) = \frac{\alpha_r}{r!}x^r e^{\gamma x} \prod_{\nu=1}^{\infty} (1 + \gamma_\nu x) \quad (\text{I})$$

where

$$\alpha_\nu \neq 0, \gamma \geq 0, \text{ and } \gamma_\nu \geq 0.$$

Similarly, an entire function

$$\Psi(x) = \beta_0 + \frac{\beta_1}{1!}x + \frac{\beta_2}{2!}x^2 + \dots$$

is a *function of type II* if its zeroes are all real and it can be written as a product of the form

$$\Psi(x) = \frac{\beta_r}{r!}x^r e^{-\gamma x^2 + \delta x} \prod_{\nu=1}^{\infty} (1 + \delta_\nu x) e^{-\delta_\nu x} \quad (\beta_r \neq 0)$$

where $\delta, \delta_1, \delta_2, \dots$ are reals, and γ is a non-negative real.

Obviously, a function of type I is also of type II.

These two classes of functions are characterized by the following remarkable theorems 1 and 2:

Theorem 1. *The power series*

$$\Phi(x) = \alpha_0 + \frac{\alpha_1}{1!}x + \frac{\alpha_2}{2!}x^2 + \dots$$

with real coefficients is an entire function of type I if and only if there exists a sequence of polynomials

$$\Phi_n(x) = \alpha_{n0} + \frac{\alpha_{n1}}{1!}x + \frac{\alpha_{n2}}{2!}x^2 + \dots \quad (n = 1, 2, \dots),$$

each with only real zeroes of the same sign, which converges uniformly to $\Phi(x)$ on every closed disc $|x| \leq \rho$.

In the proof of this theorem, without loss of generality, we may assume that the zeroes of the polynomials $\Phi_n(x)$ (or the function $\Phi(x)$) are ≤ 0 and the coefficients $\alpha_{n\nu}$ (or α_ν) are ≥ 0 .

Suppose that the polynomials $\Phi_n(x)$ converge uniformly to $\Phi(x)$ for $|x| \leq \rho$. Then we know that

$$\lim_{n \rightarrow \infty} \alpha_{n\nu} = \alpha_\nu. \quad (\nu = 0, 1, 2, \dots)$$

For any positive integer m , the roots of the equation

$$(1+x)^m = \sum_{\mu=0}^m \binom{m}{\mu} x^\mu = 0$$

are real and negative, then by the composition theorem^a from A, the equation

$$\sum_{\mu=0}^m \mu! \frac{\alpha_{n\mu}}{\mu!} \binom{m}{\mu} x^\mu = 0 \tag{6}$$

has only real roots, and by our assumption on the polynomials $\Phi_n(x)$, the non-zero roots are negative. Letting n tend to infinity in (6), we obtain the equation

$$F_m(x) = \alpha_0 + \binom{m}{1} \alpha_1 x + \binom{m}{2} \alpha_2 x^2 + \dots + \alpha_m x^m = 0,$$

whose roots are also all real and ≤ 0 .

If all terms of the sequence $\alpha_0, \alpha_1, \alpha_2, \dots$ vanish, then $\Phi(x)$ is certainly a function of type I. Otherwise, let α_r be the first non-zero coefficient of $\Phi(x)$. For $m > r$, let

$$F_m(x) = \binom{m}{r} \alpha_r x^r \prod_{\mu=1}^{m-r} (1 + \gamma_{m\mu} x),$$

where $\gamma_{m\mu}$ are non-negative reals. Then

$$\binom{m}{r+1} \alpha_{r+1} = \binom{m}{r} \alpha_r \sum_{\mu=1}^{m-r} \gamma_{m\mu} \tag{7}$$

and

$$\alpha_m = \binom{m}{r} \alpha_r \prod_{\mu=1}^{m-r} \gamma_{m\mu}.$$

The AM-GM inequality implies

$$\frac{\alpha_m}{\binom{m}{r} \alpha_r} \leq \left\{ \frac{\binom{m}{r+1} \alpha_{r+1}}{(m-r) \binom{m}{r} \alpha_r} \right\}^{m-r}$$

or, equivalently,

$$\alpha_m \leq \binom{m}{r} \alpha_r \left(\frac{\alpha_{r+1}}{(r+1) \alpha_r} \right)^{m-r}. \quad (8)$$

Thus

$$\frac{\alpha_m}{m!} \leq \frac{1}{(m-r)!} \frac{\alpha_r}{r!} \left(\frac{\alpha_{r+1}}{(r+1) \alpha_r} \right)^{m-r}.$$

So the series $\Phi(x)$ is absolutely convergent. By Hadamard Theory, it also follows that $\Phi(x)$ is of genus at most 1.

Now, it is easy to see that the polynomials

$$\begin{aligned} F_m\left(\frac{x}{m}\right) &= \alpha_0 + \frac{\alpha_1}{1!}x + \frac{\alpha_2}{2!}\left(1 - \frac{1}{m}\right)x + \frac{\alpha_3}{3!}\left(1 - \frac{1}{m}\right)\left(1 - \frac{2}{m}\right)x^2 + \dots \\ &= \frac{\alpha_r}{r!}\left(1 - \frac{1}{m}\right) \dots \left(1 - \frac{r-1}{m}\right) x^r \prod_{\mu=1}^{m-r} \left(1 + \frac{\gamma_{m\mu}}{m}x\right) \end{aligned}$$

converge uniformly to $\Phi(x)$ on every bounded region. This means that the zeroes

$$-\frac{1}{\gamma_1}, -\frac{1}{\gamma_2}, -\frac{1}{\gamma_3}, \dots$$

of the function $\Phi(x)$ (if they exist) are all real and have the same sign as the numbers $-\frac{m}{\gamma_{m\mu}}$.

We may write (7) as

$$\sum_{\mu=1}^{m-r} \frac{\gamma_{m\mu}}{m} = \frac{m-r}{m} \frac{\alpha_{r+1}}{(r+1) \alpha_r}.$$

It is easy to see that $\sum_{\mu=1}^{\infty} \gamma_{\mu}$ converges and

$$\sum_{\mu=1}^{\infty} \gamma_{\mu} \leq \frac{\alpha_{r+1}}{(r+1) \alpha_r}. \quad (9)$$

Therefore, we conclude that

$$\Phi(x) = \frac{\alpha_r}{r!} x^r + \frac{\alpha_{r+1}}{(r+1)!} x^{r+1} + \dots = \frac{\alpha_r}{r!} x^r e^{\gamma x} \prod_{\nu=1}^{\infty} (1 + \gamma_{\nu} x)$$

and

$$\frac{\alpha_{r+1}}{(r+1)!} = \frac{\alpha_r}{r!} \left(\gamma + \sum_{\nu=1}^{\infty} \gamma_{\nu} \right)$$

which means that γ is not negative because of (9).

Conversely, it is very easy to see that for every function of type I (with non-positive zeroes) a required polynomial sequence can be specified. If

$$\Phi(x) = \frac{\alpha_r}{r!} x^r e^{\gamma x} \prod_{\nu=1}^{\infty} (1 + \gamma_{\nu} x) \quad (\gamma, \gamma_{\nu} \geq 0)$$

is such a function, we set

$$\Phi_n(x) = \frac{\alpha_r}{r!} x^r \left(1 + \frac{\gamma x}{n} \right)^n \prod_{\nu=1}^n (1 + \gamma_{\nu} x).$$

This defines a sequence of polynomials with only non-positive zeroes that converge uniformly to $\Phi(x)$ on every bounded region.

Theorem 2. *The power series*

$$\Psi(x) = \beta_0 + \frac{\beta_1}{1!} x + \frac{\beta_2}{2!} x^2 + \dots$$

with real coefficients is an entire function of type II if and only if there exists a sequence of polynomials

$$\Psi_n(x) = \beta_{n0} + \frac{\beta_{n1}}{1!} x + \frac{\beta_{n2}}{2!} x^2 + \dots$$

each with only real zeroes, which converges uniformly to $\Psi(x)$ on every closed disc $|x| \leq \rho$.

If the polynomials $\Psi_n(x)$ converge uniformly to $\Psi(x)$ for $|x| \leq \rho$, then in the same way as done in the proof of Theorem 1, we conclude from the composition theorem that the equation

$$G_m(x) = \beta_0 + \binom{m}{1} \beta_1 x + \binom{m}{2} \beta_2 x^2 + \dots + \beta_m x^m = 0,$$

for each m , has only real roots, however in this case they may not have the same sign. Suppose β_r is the first non-zero coefficient of the series $\Psi(x)$, and let $m > r + 1$ and

$$G_m(x) = \binom{m}{r} \beta_r x^r \prod_{\mu=1}^{m-r} (1 + \delta_{m\mu} x).$$

Then

$$\binom{m}{r+1}^2 \beta_{r+1}^2 - 2 \binom{m}{r} \binom{m}{r+2} \beta_r \beta_{r+2} = \binom{m}{r}^2 \beta_r^2 \sum_{\mu=1}^{m-r} \delta_{m\mu}^2$$

and

$$\beta_m = \binom{m}{r} \beta_r \prod_{\mu=1}^{m-r} \delta_{m\mu}.$$

Applying the AM-GM inequality to the numbers $\delta_m \mu^2$, it follows that

$$\frac{\beta_m^2}{\binom{m}{r}^2 \beta_r^2} \leq \left\{ \frac{\binom{m}{r+1}^2 \beta_{r+1}^2 - 2 \binom{m}{r} \binom{m}{r+2} \beta_r \beta_{r+2}}{(m-r) \binom{m}{r}^2 \beta_r^2} \right\}^{m-r},$$

or more simply,

$$\beta_m^2 \leq \binom{m}{r}^2 \beta_r^2 \left\{ \frac{(m-r) \beta_{r+1}^2}{(r+1)^2 \beta_r^2} - \frac{2(m-r-1) \beta_{r+2}}{(r+1)(r+2) \beta_r} \right\}^{m-r}$$

Therefore

$$\beta_m^2 \leq m^m c^m,$$

where c is some constant (that only depends on r , β_r , β_{r+1} and β_{r+2}). Thus

$$\frac{|\beta_m|}{m!} \leq \frac{m^{m/2}}{m!} c^{m/2}, \quad (10)$$

which proves that the series $\Psi(x)$ is absolutely convergent; Hadamard theory also tells us that $\Psi(x)$ is at most of genus 2.

If we regard $\Psi_n(x)\Psi_n(-x)$ as a rational entire function of x^2 , it has only real non-negative roots⁴ and the polynomials $\Psi_n(x)\Psi_n(-x)$ converge uniformly to $\Psi(x)\Psi(-x)$ in the disc $|x^2| \leq \rho^2$, then based on Theorem 1, it follows that $\Psi(x)\Psi(-x)$ regarded as a function of x^2 , is a function of type 1 with only non-negative zeroes.

It follows, firstly, that the zeroes of $\Psi(x)$ are real, and secondly, that the sum of the squares of the reciprocal zeroes is convergent, and finally, that in the product expansion

$$\Psi(x) = \frac{\beta_r}{r!} x^r e^{-\gamma x^2 + \delta x} \prod_{\nu=1}^{\infty} (1 + \delta_{\nu} x) e^{-\delta_{\nu} x} \quad (II)$$

the number γ because we have

$$(-1)^r \Psi(x)\Psi(-x) = \frac{\beta_r^2}{r!^2} x^{2r} e^{-2\gamma x^2} \prod_{\nu=1}^{\infty} (1 - \delta_{\nu}^2 x^2).$$

Conversely, if $\Psi(x)$ is such an entire function (of the form (II)), we set

$$\delta - \delta_1 - \delta_2 - \dots - \delta_n = d_n$$

⁴Its roots are the squares of the roots of $\Psi_n(x)$.

and let m_n be a (the?) positive integer such that for $|x| < n$

$$\left| 1 - \frac{\left(1 + \frac{d_n x}{m_n}\right)^{m_n}}{e^{d_n x}} \right| < \frac{1}{n}$$

The polynomials defined by

$$\Psi_n(x) = \frac{\beta_r}{r!} x^r \left(1 - \frac{\gamma x^2}{n}\right)^n \left(1 + \frac{d_n x}{m_n}\right)^{m_n} \prod_{\nu=1}^n (1 + \delta_\nu x),$$

obviously have real zeroes, and it is easily seen that the following limit is uniform on every bounded region.

$$\lim_{n \rightarrow \infty} \Psi_n(x) = \Psi(x).$$

This completes the proof of Theorem 2.

Laguerre⁵ has already proved that a sequence of polynomials with only real roots of the same sign, which is uniformly convergent in every finite region, always represents an entire function of the first type. He has also stated the analogous theorem for polynomials with only real zeros without proof. That in both cases the uniform convergence in a disc $|x| \leq \rho$, however small, is already sufficient, was first shown by Pólya⁶. The proof given here, which is primarily based on the composition theorem, seems to us to be of interest because of its algebraic character.

To give an immediate application of these results, we mention the following. Let

$$\begin{vmatrix} a_{11} & a_{12} & \dots \\ a_{21} & a_{22} & \dots \\ \vdots & \vdots & \ddots \end{vmatrix}$$

be an infinite symmetric matrix with real entries. Set

$$\Delta_n(x) = \begin{vmatrix} 1 - a_{11}x & -a_{12}x & \dots & -a_{1n}x \\ -a_{21}x & 1 - a_{22}x & \dots & -a_{2n}x \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1}x & -a_{n2}x & \dots & 1 - a_{nn}x \end{vmatrix}.$$

If then in a disc $|x| \leq \rho$, however small, $\lim_{n \rightarrow \infty} \Delta_n(x) = A(x)$ uniformly, then

$$\Delta(x) = \begin{vmatrix} 1 - a_{11}x & -a_{12}x & \dots \\ -a_{21}x & 1 - a_{22}x & \dots \\ \vdots & \vdots & \ddots \end{vmatrix}$$

⁵Oeuvres de Laguerre, vol. I, pp. 174-177.

⁶Über Annäherung durch Polynome mit lauter reellen Wurzeln, Rend. del Circ. mat. di Palermo, vol. 36 (1913), p. 279. — Further theorems can be found in the work: E. Lindwart and G. Pólya, Über einen Zusammenhang zwischen der Konvergenz von Polynomfolgen und der Verteilung ihrer Wurzel, Rend. del Cic. mat. di Palermo, vol. 37 (1914).

is an entire function of type II.

For completeness, we add the following remark:

If

$$\Omega_n(x) = \gamma_{n0} + \frac{\gamma_{n1}}{1!}x + \frac{\gamma_{n2}}{2!}x^2 + \dots$$

are polynomials whose zeroes lie above or on the real axis, and if in a circle $|x| \leq \rho$, however small,

$$\lim_{n \rightarrow \infty} \Omega_n(x) = \Omega(x) = \gamma_0 + \frac{\gamma_1}{1!}x + \frac{\gamma_2}{2!}x^2 + \dots,$$

uniformly, then $\Omega(x)$ is an entire function of at most genus 2.

For if

$$\gamma_{n\nu} = \gamma'_{n\nu} + i\gamma''_{n\nu}, \quad \gamma_\nu = \gamma'_\nu + i\gamma''_\nu$$

($\gamma'_{n\nu}$, $\gamma''_{n\nu}$, γ'_ν , γ''_ν real) and set

$$\begin{aligned} \Omega_n^{(1)}(x) &= \sum_{\nu=0}^{\infty} \frac{\gamma'_{n\nu}}{\nu!} x^\nu, & \Omega_n^{(2)}(x) &= \sum_{\nu=0}^{\infty} \frac{\gamma''_{n\nu}}{\nu!} x^\nu, \\ \Omega^{(1)}(x) &= \sum_{\nu=0}^{\infty} \frac{\gamma'_\nu}{\nu!} x^\nu, & \Omega^{(2)}(x) &= \sum_{\nu=0}^{\infty} \frac{\gamma''_\nu}{\nu!} x^\nu, \end{aligned}$$

then uniformly for $|x| \leq \rho$

$$\lim_{n \rightarrow \infty} \Omega_n^{(1)}(x) = \Omega^{(1)}(x), \quad \lim_{n \rightarrow \infty} \Omega_n^{(2)}(x) = \Omega^{(2)}(x).$$

Now, using a well-known theorem of Biehler⁷, the zeroes of the polynomials $\Omega_n^{(1)}(x)$ and $\Omega_n^{(2)}(x)$, due to the assumption made about $\Omega_n(x)$, are all real. According to inequality (10) found in the proof of Theorem II, there exist positive constants c' and c'' such that

$$\frac{|\gamma'_m|}{m!} \leq \frac{m^{m/2}}{m!} c'^m, \quad \frac{|\gamma''_m|}{m!} \leq \frac{m^{m/2}}{m!} c''^m.$$

From this, however, follows the above-mentioned assertion about the function $\Omega(x)$ ⁸.

3 Algebraic criteria for multiplier sequences

Theorem 1. *A sequence of real numbers*

$$\alpha_0, \alpha_1, \alpha_2, \dots \tag{A}$$

⁷cf. *Netto*, Vorlesungen über Algebra, vol. I, pp. 236-237.

⁸cf. *Lindwart and Pólya*, a. a. O.

is a multiplier sequence of the first kind if and only if the equations

$$\alpha_0 + \binom{n}{1}\alpha_1x + \binom{n}{2}\alpha_2x^2 + \cdots + \alpha_nx^n = 0 \quad (n = 1, 2, 3, \dots) \quad (11)$$

have only real roots of the same sign.

For if (A) is a multiplier sequence of the first kind, then from the fact that the roots of the equation

$$1 + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + x^n = 0$$

are all real, it follows immediately from the definition of multiplier sequences of the first kind that equation (11) also only has real roots.

From Theorem 1 of §1, it further follows that these roots are all of the same sign.

If, conversely, it is known that each of the equations (11) has all real roots of the same sign, then let

$$a_0 + a_1x + a_2x^2 + \cdots + a_mx^m = 0$$

be any equation with only real roots. From the composition theorem, it follows that for each n the equation

$$a_0\alpha_0 + 1!a_1\alpha_1\binom{n}{1}x + \cdots + m!a_m\alpha_m\binom{n}{m}x^m = 0$$

has only real roots. If one replaces x by $\frac{x}{n}$, one obtains the equation

$$\begin{aligned} \alpha_0a_0 + \alpha_1a_1x + \alpha_2a_2\left(1 - \frac{1}{n}\right)x + \cdots \\ \alpha_ma_m\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{m-1}{n}\right) = 0, \end{aligned}$$

whose roots are obviously also all real. Since this holds for each n , it follows by taking the limit that the equation

$$\alpha_0a_0 + \alpha_1a_1x + \alpha_2a_2x^2 + \cdots + \alpha_ma_mx^m = 0$$

has only real roots. — With this, Theorem 1 is completely proved.

Theorem 2. A sequence of real numbers

$$\beta_0, \beta_1, \beta_2, \dots \quad (B)$$

is a multiplier sequence of the second kind if and only if the equations

$$\beta_0 + \binom{n}{1}\beta_1x + \binom{n}{2}\beta_2x^2 + \cdots + \beta_nx^n = 0 \quad (n = 1, 2, 3, \dots) \quad (12)$$

have only real roots.

That this condition is necessary is immediately clear. But is also sufficient. Let

$$b_0 + b_1x + b_2x^2 + \cdots + b_mx^m = 0$$

be an equation with all real roots of the same sign. By composing this equation with (12), which, by assumption, should have all real roots, it follows again from the composition theorem just used that the equation

$$\beta_0b_0 + 1!\binom{n}{1}\beta_1b_1x + 2!\binom{n}{2}\beta_2b_2x^2 + \cdots + m!\binom{n}{m}\beta_mb_mx^m = 0 \quad (n = 1, 2, 3, \dots)$$

has all real roots, and now one concludes exactly as before that the equation

$$\beta_0b_0 + \beta_1b_1x + \beta_2b_2x^2 + \cdots + \beta_mb_mx^m = 0$$

has all roots real.

Theorem 3. *A multiplier sequence of the second kind*

$$\beta_0, \beta_1, \beta_2, \dots$$

is also a multiplier sequence of the first kind if the non-zero terms of the sequence form an uninterrupted sequence of numbers, either of the same sign or with alternating signs.

For if, and only if, this condition is fulfilled are all the real roots of (12) of the same sign.

The argument for Theorem 1 and 2 also proves an interesting property of multiplier sequences of the second kind.

Theorem 4. *Let*

$$\beta_0, \beta_1, \beta_2, \dots$$

be a multiplier sequence of the second kind. If the numbers

$$\beta_l, \beta_{l+1}, \beta_{l+2}, \dots, \beta_{l+m}$$

are non-zero and either have the same sign or have alternating signs, then if

$$a_0 + a_1x + a_2x^2 + \cdots + a_mx^m = 0$$

is any equation with only real roots, the equation

$$\beta_la_0 + \beta_{l+1}a_1x + \beta_{l+2}a_2x^2 + \cdots + \beta_{l+m}a_mx^m = 0$$

also has only real roots.

Based on Theorem 5 of §1, the sequence

$$\beta_l, \beta_{l+1}, \beta_{l+2}, \dots$$

is a multiplier sequence of the second kind. Therefore, each of the equations

$$\beta_l + \binom{n}{1}\beta_{l+1}x + \binom{n}{2}\beta_{l+2}x^2 + \dots + \beta_{l+m}x^m = 0$$

has all roots real. Consequently, the roots of the equation

$$\beta_l a_0 + 1! \binom{n}{1} \beta_{l+1} a_1 x + \dots + m! \binom{n}{m} \beta_{l+m} a_m x^m = 0$$

are all real. If one replaces x by $\frac{x}{n}$ and by letting n approach infinity, one obtain Theorem 4.

We add one more remark, which offers a new perspective on multiplier sequences and the constantly used composition theorem.

We wish to determine all sequences

$$\gamma_0, \gamma_1, \gamma_2, \dots, \gamma_\nu, \dots \quad (13)$$

that have the following property: if the two equations

$$\begin{aligned} f(x) &= a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m = 0, \\ g(x) &= b_0 + b_1 x + b_2 x^2 + \dots + b_n x^n = 0 \end{aligned}$$

has only real roots and those of the second equation are of the same sign, then, if k is the smaller of the two numbers m and n , the equation

$$\gamma_0 a_0 b_0 + \gamma_1 a_1 b_1 x + \gamma_2 a_2 b_2 x^2 + \dots + \gamma_k a_k b_k x^k = 0 \quad (14)$$

has only real roots.

The key: the sequence (13) has the desired property if and only if the sequence

$$\gamma_0, \frac{\gamma_1}{1!}, \frac{\gamma_2}{2!}, \dots, \frac{\gamma_\nu}{\nu!}, \dots \quad (15)$$

is a multiplier sequence of the first kind.

If (15) is a multiplier sequence of the first kind, then by the composition theorem, the equation

$$a_0 b_0 + 1! a_1 b_1 x + \dots + k! a_k b_k x^k = 0$$

and consequently also the equation

$$\gamma_0 a_0 b_0 + \frac{\gamma_1}{1!} 1! a_1 b_1 x + \dots + \frac{\gamma_k}{k!} k! a_k b_k x^k = 0$$

has all roots real.

If, conversely, we know that (13) has the desired property, then we choose $f(x)$ arbitrarily and $g(x) = \left(1 + \frac{x}{n}\right)^n$. Then equation (14), whose roots are real, for $n \geq m$ becomes

$$\gamma_0 a_0 + \gamma_1 a_1 \frac{x}{n} + \cdots + \gamma_m a_m \frac{x^m}{n^m} = 0.$$

If we let n tend to infinity, we obtain

$$\gamma_0 a_0 + \frac{\gamma_1}{1!} a_1 x + \frac{\gamma_2}{2!} a_2 x^2 + \cdots + \frac{\gamma_m}{m!} a_m x^m = 0.$$

Since this equation has all roots real simultaneously with $f(x) = 0$, however may the roots of $f(x) = 0$ be chosen, the sequence (15) is a multiplier sequence of the first kind.

This result can also be stated as follows: All “multiplier sequences” for the special class of equations

$$a_0 b_0 + a_1 b_1 x + a_2 b_2 x^2 + \cdots + a_k b_k x^k$$

with only real roots, to which the Malo theorem refers, are obtained by composing all multiplier sequences of the first kind with the sequence

$$0!, 1!, 2!, 3!, \dots, \nu!, \dots$$

4 Transcendental criteria for multiplier sequences

Theorem 1. *A sequence of real numbers*

$$\alpha_0, \alpha_1, \alpha_2, \dots \tag{A}$$

is a multiplier sequence of the first kind if and only if the power series

$$\Phi(x) = \alpha_0 + \frac{\alpha_1}{1!} x + \frac{\alpha_2}{2!} x^2 + \dots$$

converges absolutely and represents an entire function of type I⁹.

If (A) is a multiplier sequence of the first kind, then the equations (11) each have only real roots of the same sign. From this, however, as argued in the proof of Theorem 1 of §2 (cf. (7), (8), (9)), it follows that the series $\Phi(x)$ is absolutely convergent and is an entire function of type I.

⁹The series $\Phi(x)$ has been studied by *Laguerre* (Oeuvres, book 1, p. 177 and pp. 202-204) and *Hurwitz*, Über die Wurzeln einiger transzendenter Gleichungen (Hamb. Mitt. II, 1890, p. 25) for specific multiplier sequences and it has been established that in these cases the series represents an entire function of type I. See also *Jensen*, Recherches sur la théorie des équations, Acta Math., vol. 36, 1912-13, pp. 181-195.

If, conversely, $\Phi(x)$ is an entire function of type I, then, using the theorem just mentioned, there exists a sequence of polynomials

$$\Phi_n(x) = \alpha_{n0} + \frac{\alpha_{n1}}{1!}x + \frac{\alpha_{n2}}{2!}x^2 + \dots \quad (n = 1, 2, 3, \dots)$$

whose roots are all real and of the same sign, so that in every compact region

$$\lim_{n \rightarrow \infty} \Phi_n(x) = \Phi(x)$$

uniformly and so

$$\lim_{n \rightarrow \infty} \alpha_{n\nu} = \alpha_\nu \quad (\nu = 0, 1, 2, 3, \dots) \quad (16)$$

holds.

Suppose now that

$$a_0 + a_1x + a_2x^2 + \dots + a_mx^m = 0$$

is any equation with only real roots, then from the composition theorem it follows that the equations

$$\alpha_{n0}a_0 + \alpha_{n1}a_1x + \alpha_{n2}a_2x^2 + \dots + \alpha_{nm}a_mx^m = 0 \quad (n = 1, 2, 3, \dots)$$

have only real roots. Thus, because of (16), the roots of the equation

$$\alpha_0a_0 + \alpha_1a_1x + \alpha_2a_2x^2 + \dots + \alpha_ma_mx^m = 0$$

are all real.

Theorem 2. *A sequence of real numbers*

$$\beta_0, \beta_1, \beta_2, \dots$$

is a multiplier sequence of the first kind if and only if the power series

$$\Psi(x) = \beta_0 + \frac{\beta_1}{1!}x + \frac{\beta_2}{2!}x^2 + \dots$$

converges absolutely and represents an entire function of type II.

The proof is very similar to that of Theorem 1.

5 Examples

Using Theorem 1 and 2 of the preceding §4, we shall obtain some noteworthy multiplier sequences.

1. Since for every positive integer r

$$x^r e^x = r! \frac{x^r}{r!} + \frac{(r+1)!}{1!} \cdot \frac{x^{r+1}}{(r+1)!} + \frac{(r+2)!}{2!} \cdot \frac{x^{r+2}}{(r+2)!} + \dots$$

is an entire function of type I, the sequence

$$0, 0, \dots, 0, r!, \frac{(r+1)!}{1!}, \frac{(r+2)!}{2!}, \dots$$

is a multiplier sequence of the first kind. This is merely the fact that the r -th derivative of a polynomial with only real zeroes has the same property.

2. The function

$$e^{-\frac{x^2}{2}} = 1 - \frac{x^2}{2!} + 1 \cdot 3 \frac{x^4}{4!} - 1 \cdot 3 \cdot 5 \frac{x^6}{6!} + \dots$$

is of type II, therefore the sequence

$$1, 0, -1, 0, 1 \cdot 3, 0, -1 \cdot 3 \cdot 5, 0, \dots$$

is a multiplier sequence of the second kind.

3. If λ and θ are two real numbers, then, a simple calculation shows,

$$F(x) = e^{x \cos \vartheta} \cos(\lambda + x \sin \vartheta) = \sum_{\nu=0}^{\infty} \cos(\lambda + \nu \vartheta) \frac{x^\nu}{\nu!}.$$

Since the function $F(x)$ on the left is of type II, the sequence (3) is a multiplier sequence of the second kind¹⁰. Of particular interest are the special cases $F(x) = \cos(x)$ and $F(x) = \sin(x)$, which show that both sequences

$$\begin{array}{cccccccc} 1, & 0, & -1, & 0, & 1, & 0, & -1, & \dots \\ 0, & 1, & 0, & -1, & 0, & 1, & 0, & \dots \end{array}$$

are remarkably multiplier sequences of the second kind.

If

$$a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m = 0$$

is an equation with only real roots (they do not need to be of the same sign), and if the $m+1$ numbers

$$\lambda, \lambda + \vartheta, \lambda + 2\vartheta, \dots, \lambda + m\vartheta$$

all lie in the interval $\left(-\frac{\pi}{2}, +\frac{\pi}{2}\right)$, then it follows from Theorem 4 of §3 that the equation

$$a_0 \cos \lambda + a_1 \cos(\lambda + \vartheta) + \dots + a_m \cos(\lambda + m\vartheta) x^m = 0$$

has only real roots.

¹⁰Cf. Introduction. Laguerre (Ouevres, vol. 1, pp. 204-205) deduces this from Biehler's theorem.

4. Since the product of two functions of type I (or II) is again a function of the same type, we have the following rule:

If

$$\begin{array}{ccccccc} \gamma_0, & \gamma_1, & \gamma_2, & \dots, & \gamma_\nu, & \dots \\ \gamma'_0, & \gamma'_1, & \gamma'_2, & \dots, & \gamma'_\nu, & \dots \end{array}$$

are two multiplier sequences of the first (or second) kind, then the numbers

$$\gamma''_\nu = \gamma_0 \gamma'_\nu + \binom{\nu}{1} \gamma_1 \gamma'_{\nu-1} + \binom{\nu}{2} \gamma_2 \gamma'_{\nu-2} + \dots + \gamma_\nu \gamma'_0 \quad (\nu = 0, 1, 2, 3, \dots)$$

also form a multiplier sequence of the same kind.

5. We have m real numbers

$$\gamma_1, \gamma_2, \dots, \gamma_m.$$

We form, for a polynomial $g(x)$ of arbitrary degree, the equation

$$G(x) = g(x) + \gamma_1 \frac{xg'(x)}{1!} + \gamma_2 \frac{x^2g''(x)}{2!} + \dots + \gamma_m \frac{x^m g^{(m)}(x)}{m!} = 0.$$

If this equation is to have only real roots for every polynomial $g(x)$ whose roots are all real, then the necessary and sufficient condition is: the equation

$$1 + \frac{\gamma_1}{1!}x + \frac{\gamma_2}{2!}x^2 + \dots + \frac{\gamma_m}{m!}x^m = 0 \quad (17)$$

must have only negative real roots.

If, however, one only requires that the equation

$$G(x) = 0$$

is to have only real roots for every polynomial $g(x)$ whose roots are all real and of the same sign, then the necessary and sufficient condition is: the equation (17) must have only real roots¹¹.

For if one sets

$$g(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n,$$

then

$$G(x) = \alpha_0 a_0 + \alpha_1 a_1 x + \dots + \alpha_m a_m x^m;$$

where

$$\alpha_\nu = 1 + \binom{\nu}{1} \gamma_1 + \binom{\nu}{2} \gamma_2 + \binom{\nu}{3} \gamma_3 + \dots + \binom{\nu}{m} \gamma_m. \quad (\nu = 0, 1, 2, \dots)$$

¹¹The following proof may also be used to prove an analogous statement for infinitely instead of finitely many $\gamma_1, \gamma_2, \dots, \gamma_m$.

We thus only have to establish the conditions under which the numbers α_ν form a multiplier sequence of the first or second kind, or equivalently, the conditions under which

$$\Omega(x) = \sum_0^{\infty} \frac{\alpha_\nu}{\nu!} x^\nu$$

is an entire function of type I or II. Both assertions followed by noticing that

$$\Omega(x) = e^x \left(1 + \frac{\gamma_1}{1!} x + \frac{\gamma_2}{2!} x^2 + \cdots + \frac{\gamma_m}{m!} x^m \right).$$

If one sets, for example, $m = 2$ and chooses for

$$1 + \frac{\gamma_1}{1!} x + \frac{\gamma_2}{2!} x^2$$

successively the three functions

$$(1+x)^2, \quad 1+ax+ax^2 \ (a \geq 4), \quad (1-x)^2,$$

then one recognizes that the numbers

$$1+v+v^2, \quad 1+av^2 \ (a \geq 4) \quad (\nu = 0, 1, 2, 3, \dots)$$

are multiplier sequences of the first kind and the numbers

$$1-3\nu+\nu^2 \quad (\nu = 0, 1, 2, 3, \dots)$$

are a multiplier sequence of the second kind. These examples, while simple, seem to be new.

6 Applications to Entire Transcendental Functions of Genus 0 and 1

The results of §§2 and 4 show that all the functions, which we referred to as functions of type I and II, can be characterized in three different ways. Firstly, by the form of their product expansion (cf. §2, formulas (I) and (II)). Secondly, they are the only entire functions that can be approximation by polynomials with only real zeroes of the same sign or any sign. And finally, by the fact that the values of their successive derivatives at the point $x = 0$ yield a multiplier sequence of the first or second kind.

This latter relationship of functions of type I and II to algebraic equations with only real roots provides a peculiar method for studying these functions, which will be used below to prove some theorems. Since our functions of type II includes all entire functions of genus zero and one whose coefficients and zeroes are real, we will find some well-known theorems about functions of genus 0 and 1 in the following.

Theorem 1. *If*

$$\Omega(x) = \gamma_0 + \frac{\gamma_1}{1!}x + \frac{\gamma_2}{2!} + \dots$$

is an entire function of type I (or II), then its derivative

$$\Omega'(x) = \gamma_1 + \frac{\gamma_2}{1!}x + \frac{\gamma_3}{2!}x^2 + \dots$$

is of the same type¹².

Since the sequence

$$\gamma_0, \gamma_1, \gamma_2, \dots$$

is a multiplier sequence of the first (or second) kind, it follows from Theorem 5 of §1 that the sequence

$$\gamma_1, \gamma_2, \gamma_3, \dots$$

is also a multiplier sequence of the first (or second) kind.

The equation

$$\frac{d^n e^{-\frac{x^2}{2}}}{dx^n} = (-1)^n U_n(x) e^{-\frac{x^2}{2}},$$

defines $U_n(x)$ which is the so-called Hermite polynomial

$$U_n(x) = x^n - \binom{n}{2} 1x^{n-2} + \binom{n}{4} 1 \cdot 3x^{n-4} - \dots$$

Since $e^{-\frac{x^2}{2}}$ is a function of type II, the well-known fact that $U_n(x) = 0$ has only real roots follows from our theorem.

A more general consequence may also be mentioned: if

$$\Psi(x) = \alpha_0 e^{-\gamma x^2 + \delta x} \prod_{\nu=1}^{\infty} (1 + \delta_{\nu} x) e^{-\delta_{\nu} x}$$

is a function of type II, then according to Theorem 1,

$$\Psi'(x) = \alpha_1 e^{-\gamma' x^2 + \delta' x} \prod_{\nu=1}^{\infty} (1 + \delta'_{\nu} x) e^{-\delta'_{\nu} x}$$

is a function of the same type. We now claim that $\gamma' \geq \gamma$. Let $\psi_1(x), \psi_2(x), \dots$ be polynomials with all real roots, which converge uniformly to the entire function of type II $e^{\gamma x} \Psi(x)$ uniformly on $|x| \leq \rho$. Thus,

$$\lim_{n \rightarrow \infty} e^{-\gamma x^2} \psi_n(x) = \Psi(x).$$

¹²Cf. *Borel*, Leçons sur les fonctions entières, p. 32.

From this, it follows for $|x| < \rho$

$$\lim_{n \rightarrow \infty} e^{-\gamma x^2} (\psi'_n(x) - 2\gamma x \psi_n(x)) = \Psi'(x).$$

The polynomials $\psi'_n - 2\gamma x \psi_n$ have, however, only real zeroes, since $e^{-\gamma x} \psi_n(x)$ is also a function of type II. Therefore, according to Theorem 2 of §2

$$\lim(\psi'_n - 2\gamma x \psi_n) = e^{-\gamma'' x} \Psi_1(x),$$

where $\Psi_1(x)$ is a function of genus one and γ'' is a non-negative real constant. Thus, $\gamma' = \gamma + \gamma'' \geq \gamma$.

In a very similar way, one can obtain an analogous result for functions of type I.

Theorem 2. *For every function*

$$\Psi(x) = \beta_0 + \frac{\beta_1}{1!}x + \frac{\beta_2}{2!}x^2 + \dots$$

of type II, the inequalities

$$\beta_\nu^2 - \beta_{\nu-1}\beta_{\nu+1} \geq 0$$

hold. If β_ν and $\beta_{\nu+1}$ are both zero, then either $\Psi(x)$ is a polynomial, at most of degree $\nu - 1$, or $x = 0$ is a zero with multiplicity at least $r + 2$.¹³

Since

$$\beta_0, \beta_1, \beta_2, \dots$$

is a multiplier sequence of the second kind, this theorem is similar to Theorems 2 and 3 of §1.

Theorem 3. *An entire function of type II, whose non-zero coefficients are positive, is a function of type I of the form*

$$e^{\gamma x} \prod_{\nu=1}^{\infty} (1 + \gamma_\nu x),$$

where $\gamma, \gamma_1, \gamma_2, \dots$ are non-negative numbers.¹⁴

The proof follows immediately from Theorem 3 of §3.

Based on this theorem, one recognises, for example, that among the coefficients of the powers of x in the expansion of

$$\frac{1}{\Gamma(x)} = x e^{Cx} \prod_{n=1}^{\infty} \left(1 + \frac{x}{n}\right) e^{-\frac{x}{n}}$$

negative numbers must occur. This function, whose zeroes are all negative, is indeed of type II, but not of type I.

¹³Cf. Borel, a. a. O., p. 35.

¹⁴This theorem seems to be new. Cf. Borel, a. a. O., p. 36.

Theorem 4. *A power series*

$$\gamma_0 + \frac{\gamma_1}{1!}x + \frac{\gamma_2}{2!}x^2 + \dots \quad (18)$$

is an entire function of type II (or I) if and only if the equations

$$\gamma_0 x^n + \binom{n}{1} \gamma_1 x^{n-1} + \binom{n}{2} \gamma_2 x^{n-2} + \dots + \gamma_n = 0 \quad (19)$$

for each n have only real roots (only real roots of the same sign).

This theorem only states the algebraic criteria for multiplier sequences given in §3 are equivalent to the trascendental criteria of §4.

Assuming that the power series (18) is absolutely convergent and represents an entire function of genus 0 or 1, Mr. Jensen has already shown that the reality of all roots of the equations (19) is a necessary and sufficient condition for the reality of the zeroes of the function (18).¹⁵

If one chooses in particular for (18) the function $e^{-\frac{x^2}{2}}$, then the expressions on the left in (19) are the previously mentioned Hermite polynomials

$$U_n(x) = x^n - \binom{n}{2} x^{n-2} + \binom{n}{4} 1 \cdot 3 x^{n-4} - \dots$$

Thus, a new proof for the reality of the zeroes of these polynomials is obtained.

Theorem 5. *If the function*

$$\alpha_0 + \frac{\alpha_1}{1!}x + \frac{\alpha_2}{2!}x^2 + \dots$$

is an entire function of type I and

$$\beta_0 + \frac{\beta_1}{1!}x + \frac{\beta_2}{2!}x^2 + \dots$$

is an entire function of type II, then the series

$$\alpha_0 \beta_0 + \frac{\alpha_1 \beta_1}{1!}x + \frac{\alpha_2 \beta_2}{2!}x^2 + \dots \quad (20)$$

is absolutely convergent and represents a function of type II.

Based on the assumptions,

$$\alpha_0, \alpha_1, \alpha_2, \dots$$

is a multiplier sequence of the first and

$$\beta_0, \beta_1, \beta_2, \dots$$

¹⁵A. a. O. p. 187. Cf. however: *Laguerre*, Oeuvres, vol. 1, p. 33 and p. 171.

is a multiplier sequence of the second kind, then according to Theorem 4 of §1,

$$\alpha_0\beta_0, \alpha_1\beta_1, \alpha_2\beta_2, \dots$$

is a multiplier sequence of the first kind.

We transferred the composition theorem used so often here to entire transcendental functions. The theorem can also be understood as follows: the infinite multiplier sequences of the first kind (for polynomials with only real roots) have the analogous property for any entire function of type II, and the infinite multiplier sequences of the second kind (for polynomials with only real roots of the same sign) have the analogous property for any entire function of type I.

If one combines the function (20) with the Bessel function corresponding to the multiplier sequence $\frac{1}{\nu!}$ (cf. A., §3)

$$I_0(2\sqrt{-x}) = 1 + \frac{x}{1!1!} + \frac{x^2}{2!2!} + \frac{x^3}{3!3!} + \dots,$$

then one also notes that the function

$$\alpha_0\beta_0 + \frac{\alpha_1\beta_1}{1!1!} + \frac{\alpha_2\beta_2}{2!2!}x^2 + \dots$$

is of type II.

This transfer of the Mido theorem (cf. A., §3) to entire transcendental functions has already been given by Mr. *Jensen* a. a. O.

Laguerre has shown: if $\Psi(x)$ is an entire function of type II with only negative zeroes and $\Phi(x)$ is a function of type I, whose product decomposition has the form

$$\Phi(x) = e^{\gamma x} \prod_{\nu=1}^{\infty} (1 + \gamma_{\nu} x) \quad (\gamma \geq 0, \gamma_{\nu} \geq 0)$$

then the numbers

$$\alpha_{\nu} = \frac{\Psi(\nu)}{\Phi(0)\Phi(1)\dots\Phi(\nu-1)}$$

form a multiplier sequence of the first kind. Thus if $\sum_0^{\infty} \beta_{\nu} \frac{x^{\nu}}{\nu!}$ is an entire function of type II, then the power series

$$\sum_0^{\infty} \frac{\beta_{\nu}}{\nu!} \frac{\Psi(\nu)}{\Phi(0)\Phi(1)\dots\Phi(\nu-1)} x^{\nu}.$$

is absolutely convergent, and represents a function of the same type. This result has essentially already been given by Mr. *Jensen* in a. a. O.

We finally prove a generalization of Theorems 1 and 2 of §2.

Theorem 6. *If a sequence of entire functions*

$$\Omega_n(x) = \gamma_{n0} + \frac{\gamma_{n1}}{1!}x + \frac{\gamma_{n2}}{2!}x^2 + \dots \quad (n = 1, 2, 3, \dots)$$

of type I (or type II) converge to

$$\Omega(x) = \gamma_0 + \frac{\gamma_1}{1!}x + \frac{\gamma_2}{2!}x^2 + \dots,$$

uniformly on a circle $|x| \leq \rho$, however small, then the series $\Omega(x)$ is absolutely convergent, and represents an entire function of type I (or type II).

If one interprets the sequences

$$\gamma_{n0}, \gamma_{n1}, \gamma_{n2}, \dots \quad (n = 1, 2, 3, \dots)$$

as multiplier sequences and notes that based on the assumptions made,

$$\lim_{n \rightarrow \infty} \gamma_{n\nu} = \gamma_\nu \quad (\nu = 0, 1, 2, 3, \dots) \quad (21)$$

holds, then the theorem is identical to the elementary limit Theorem 6 of §1.

Theorem 6 of this section can also be deduced from the results of §2 with the help of the same reasoning that leads to the theorem that the pullback of a point-set is closed.

The first approach, which makes use of multiplier sequences, however, leads immediately to the following fact: *if one only knows that functions $\Omega_n(x)$ are all of type I (or II), and that for each r equation (21) holds, then the series $\Omega(x)$ represents a function of the same type.*

Here, the functions $\Omega_n(x)$ need not converge in any region. This occurs, for example, if one sets

$$\Omega_n(x) = (nx)^n.$$

In this case, the limits γ_ν , all exist and are equal to zero; $\Omega(x)$ vanishes identically, thus it is a function of the required kind; the sequence diverges, however, for every x different from zero.

One can, however, show that if $\Omega(x)$ does not vanish identically, then from equations (21) the uniform convergence of the sequence $\Omega_n(x)$ in every compact subset follows.¹⁶

¹⁶We intend to go into more detail on the significance of multiplier sequences for algebraic equations with imaginary roots in the near future.

6.2 Notes

§1

- (a) Here is the statement and original proof of this proposition.

Proposition. *Let*

$$h(x) = c_0 + c_1x + \cdots + c_nx^n$$

be a polynomial of degree n with only real roots, and where a_0 is non-zero. If there is an r ($0 < r < n$) such that $c_r = 0$, then $c_{r-1}c_{r+1} < 0$. In particular, two consecutive coefficients of $h(x)$ cannot vanish.

- (b) This proposition is as follows. It is proven easily using Hurwitz's theorem from complex analysis.

Proposition. *For each $n \in \mathbb{N}$, let*

$$f_n(x) = a_{n0} + a_{n1}x + \cdots + a_{nm}x^m$$

be a polynomial with only real roots. If the following limit exists

$$\lim_{n \rightarrow \infty} a_{n\mu} = a_\mu$$

for each n , then the polynomial

$$f(x) = \alpha_0 + \alpha_1x + \cdots + \alpha_mx^m$$

also has only real roots.

§2

- (a) The 'composition theorem' refers to Theorem 1 of Schur's paper:

Theorem. *If the roots of the polynomial*

$$f(x) = a_0 + a_1x + \cdots + a_mx^m = 0 \quad (a_m \neq 0)$$

are all real and the roots of the polynomial

$$g(x) = b_0 + b_1x + \cdots + b_nx^n = 0 \quad (b_n \neq 0)$$

are all real and of the same sign, then the roots of the polynomial, where $k = \min(m, n)$,

$$a_0b_0 + 1!a_1b_1x + 2!a_2b_2x^2 + \cdots + k!a_kb_kx^k = 0$$

are all real. If $m \leq n$ and $a_0b_0 \neq 0$, then the roots are simple.

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